

Esercizio 2

Scrivere le tavole di verità delle funzioni:

A) $xz + w\bar{y}\bar{z} + \bar{w}\bar{y}\bar{z} + \bar{w}\bar{x}\bar{z}$

B) $cd + a\bar{b} + ac + \bar{a}\bar{c} + \bar{a}b + \bar{c}\bar{d}$

C) $xz + w\bar{y}\bar{z} + wx\bar{y} + \bar{w}xy + \bar{x}y\bar{z}$

D) $(\bar{a} + \bar{b} + c + \bar{d})(a + b + \bar{c} + d)$

A)

x	y	z	w	xz	w $\bar{y}\bar{z}$	$\bar{w}\bar{y}\bar{z}$	$\bar{w}\bar{x}\bar{z}$	A
0	0	0	0	0	0	1	1	1
0	0	0	1	0	1	0	1	1
0	0	1	0	0	0	0	0	0
0	0	1	1	0	0	0	0	0
0	1	0	0	0	0	0	1	1
0	1	0	1	0	0	0	0	0
0	1	1	0	0	0	0	0	0
0	1	1	1	0	0	0	0	0
1	0	0	0	0	0	1	0	1
1	0	0	1	0	1	0	0	1
1	0	1	0	1	0	0	0	1
1	0	1	1	1	0	0	0	1
1	1	0	0	0	0	0	0	0
1	1	0	1	0	0	0	0	0
1	1	1	0	1	0	0	0	1
1	1	1	1	1	0	0	0	1

B)

a	b	c	d	cd	a $\bar{b}$	ac	$\bar{a}\bar{c}$	$\bar{a}b$	$\bar{c}\bar{d}$	B
0	0	0	0	0	0	0	1	0	1	1
0	0	0	1	0	0	0	1	0	0	1
0	0	1	0	0	0	0	0	0	0	0
0	0	1	1	1	0	0	0	0	0	1
0	1	0	0	0	0	0	1	1	1	1
0	1	0	1	0	0	0	1	1	0	1
0	1	1	0	0	0	0	0	1	0	1
0	1	1	1	1	0	0	0	1	0	1
1	0	0	0	0	1	0	0	0	1	1
1	0	0	1	0	1	0	0	0	0	1
1	0	1	0	0	1	1	0	0	0	1
1	0	1	1	1	1	1	0	0	0	1
1	1	0	0	0	0	0	0	0	1	1
1	1	0	1	0	0	0	0	0	0	1
1	1	1	0	0	0	1	0	0	0	1
1	1	1	1	1	0	1	0	0	0	1

C)

x	y	z	w	xz	w $\bar{y}\bar{z}$	wx $\bar{y}$	$\bar{w}xy$	$\bar{x}y\bar{z}$	C
0	0	0	0	0	0	0	0	0	0
0	0	0	1	0	1	0	0	0	1
0	0	1	0	0	0	0	0	0	0
0	0	1	1	0	0	0	0	0	0
0	1	0	0	0	0	0	0	1	1
0	1	0	1	0	0	0	0	1	1
0	1	1	0	0	0	0	0	0	0
0	1	1	1	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0
1	0	0	1	0	1	1	0	0	1
1	0	1	0	1	0	1	0	0	1
1	0	1	1	1	0	1	0	0	1
1	1	0	0	0	0	0	1	0	1
1	1	0	1	0	0	0	1	0	1
1	1	1	0	1	0	0	0	0	1
1	1	1	1	1	0	0	0	0	1

D)

a	b	c	d	$\bar{a} + \bar{b} + c + \bar{d}$	$a + b + \bar{c} + d$	D
0	0	0	0	1	1	1
0	0	0	1	1	1	1
0	0	1	0	1	0	0
0	0	1	1	1	1	1
0	1	0	0	1	1	1
0	1	0	1	1	1	1
0	1	1	0	1	1	1
0	1	1	1	1	1	1
1	0	0	0	1	1	1
1	0	0	1	1	1	1
1	0	1	0	1	1	1
1	0	1	1	1	1	1
1	1	0	0	1	1	1
1	1	0	1	0	1	0
1	1	1	0	1	1	1
1	1	1	1	1	1	1

Esercizio 3

Si semplifichi tramite semplificazioni algebriche (rifacendosi alle proprietà note) le seguenti espressioni booleane:

$$A) xyz + xy + x \rightarrow x(\overbrace{yz + y + 1}^1) = x$$

$$B) x(\bar{y}z + z) \rightarrow x(z(\overbrace{\bar{y} + 1}^1)) = xz$$

2) B) $\bar{x} + x(x + \bar{y})(y + \bar{z})$

x	y	z	$\bar{x}$	$x + \bar{y}$	$x(x + \bar{y})$	$(y + \bar{z})$	$x(x + \bar{y}) \cdot (y + \bar{z})$	y
0	0	0	1	1	0	1	0	1
0	0	1	1	1	0	0	0	1
0	1	0	1	0	0	1	0	1
0	1	1	1	0	0	1	0	1
1	0	0	0	1	1	1	1	1
1	0	1	0	1	1	0	0	0
1	1	0	0	1	1	1	1	1
1	1	1	0	1	1	1	1	1

z \ y	00	01	11	10
0	1	1	1	1
1	1	1	1	0

$$y = \bar{x} + y + \bar{z} \quad \text{S.P.}$$

$$y = \bar{z} + \bar{x} + y = \bar{x} + y + \bar{z} \quad \text{P.S.}$$

Esercizio 4

Dimostrare, usando le semplificazioni algebriche, l'uguaglianza delle seguenti equazioni:

A) $\bar{a}b + \bar{b}c + ab + \bar{b}c = 1$

B) $y + \bar{x}z + x\bar{y} = x + y + z$

~~C) $\bar{x}y + \bar{y}z + xz + xy + y\bar{z} = \bar{x}y + xz + y\bar{z}$~~

~~D) $a\bar{b} + \bar{a}c\bar{d} + \bar{a}bd + \bar{a}bcd = \bar{b} + \bar{a}c\bar{d}$~~

A) $\bar{A}B + \bar{B}C + AB + \bar{B}C = 1 \Leftrightarrow B(\bar{A} + A) + \bar{B}(\bar{C} + C) = 1 \Leftrightarrow$
 $\Leftrightarrow B + \bar{B} = 1 \Leftrightarrow 1 = 1$

B) $y + \bar{x}z + x\bar{y} = x + y + z \Leftrightarrow y + x\bar{y} + \bar{x}z = x + y + z \Leftrightarrow$
 $\Leftrightarrow x + y + \bar{x}z = x + y + z \Leftrightarrow x + \bar{x}z + y = x + y + z \Leftrightarrow$
 $\Leftrightarrow x + y + z = x + y + z$

Esercizio 8

Semplificare, usando le k-map le seguenti funzioni

A) $\bar{x}\bar{z} + y\bar{z} + xyz$

B) $\bar{a}b + \bar{b}c + \bar{a}bc$

C) $\bar{a}\bar{b} + \bar{a}c + \bar{b}c + \bar{a}b\bar{c}$

x	y	z	$\bar{x}\bar{z}$	$y\bar{z}$	xyz	A
0	0	0	1	0	0	1
0	0	1	0	0	0	0
0	1	0	1	1	0	1
0	1	1	0	0	0	0
1	0	0	0	0	0	0
1	0	1	0	0	0	0
1	1	0	0	1	0	1
1	1	1	0	0	1	1

xy \ z	00	01	11	10
0	1	1	1	0
1	0	0	1	0

$y = \bar{x}\bar{z} + xy$

A	B	C	$\bar{A}B$	$\bar{B}C$	$\overline{ABC}$	B
0	0	0	0	0	1	1
0	0	1	0	1	1	1
0	1	0	1	0	1	1
0	1	1	1	0	1	1
1	0	0	0	0	1	1
1	0	1	0	1	1	1
1	1	0	0	0	1	1
1	1	1	0	0	0	0

xy \ z	00	01	11	10
0	1	1	1	1
1	1	1	0	1

$y = \bar{x} + \bar{z} + \bar{y}$

$y = \bar{x} + \bar{y} + \bar{z}$

A	B	C	$\bar{A}\bar{B}$	$\bar{B}C$	$A\bar{C}$	$\bar{A}B\bar{C}$	C
0	0	0	1	0	0	0	1
0	0	1	1	1	0	0	1
0	1	0	0	0	0	1	1
0	1	1	0	0	0	0	0
1	0	0	0	0	1	0	1
1	0	1	0	1	0	0	1
1	1	0	0	0	1	0	1
1	1	1	0	0	0	0	0

$\begin{matrix} A \\ B \\ C \end{matrix}$	00	01	11	10
0	1	1	1	1
1	1	0	0	1

$$Y = \bar{B} + C$$